MNLP cheat sheet (Monz), exam 22 Oct 2026 · L01 Scenarios, tasks, deep learning, LLMs **1** L02 Resource tiers **1** · English-first assumptions **1** · Writing systems **1** · Code pages and Unicode **1** · Codepoints and normalisation **1** · UTF-8, UTF-16, UTF-32 **2** · Bytes, breaking and other pitfalls **2** L03 Vocabulary, lexicon, short-lists **2** · Word segmentation: maximum matching **2** · BMES tagging with an HMM and Viterbi **2** · Word formation and morphemes **2** · Morphological typology **3** L04 Why subwords, FSTs and Morfessor **3** · BPE and WordPiece **3** · Unigram LM and DP segmentation **3** · SentencePiece, OOV and byte-level BPE **3** · BPE, WordPiece, Unigram LM compared **4** · BPE vs Unigram LM (Bostrom and Durrett 2020) **4** · Fertility **4** L05 Count vectors and embeddings **4** · Word2Vec: CBOW and skip-gram **4** · Negative sampling **4** · GloVe **4** · Evaluating embeddings **4**

L01 Overview

Scenarios, tasks, deep learning, LLMs

- **Multilingual** (training not language-specific): NER for many languages, parsing (universal features), classification across languages.
- **Crosslingual** (knowledge transfer): MT (make understandable), QA (extract), unlearning (manipulate), reasoning (combine).
- **Tasks**: QA returns the answer (IR: ranked documents, even a whole question chopped into keywords); **sentiment** pos/neg or fine-grained (X likes Y about Z), needs NER, sentence and document level; **MT** typically per sentence; **summarisation** one or several documents; **NER** much harder outside English.
- **Before deep learning**: one approach per application; ML generative (Bayesian) and discriminative; hand-built features (parse trees syntactic, named entities semantic) that ML only weighs.
- **Neural**: state of the art for all NLP. **+** strong performance, little or no feature engineering, few network types, so insights transfer to vision, speech. **–** needs large data, errors hard to trace, can fail spectacularly.
- **LLMs**: originated in NLP; do tasks end to end, substitute or supplement humans (annotation, LLM as a judge). Most are English-centric (Internet data); Llama 3.1, DeepSeek better on high-resource languages; dedicated multilingual (Aya-Expanse) tend to lag behind.

Traps: "unlearning combines information" (false: manipulates).

L02 Multilinguality and writing systems

Resource tiers

- **English dominance**: most data, tools, benchmarks, papers; other languages get by-products, translated English resources, less so dedicated effort. Pipelines assume closed vocabularies; language independence usually untested.
- **Coverage** follows commercial interest, military relevance.
- **High resource**: large labelled and unlabelled data, many tools: English, Mandarin, [big gap] French, Spanish, Arabic, Russian.
- **Medium**: limited labelled, reasonable unlabelled, some tools: German, Italian, Dutch, Korean, Japanese.
- **Low**: no labelled, limited unlabelled, no tools: remaining ≈7,000.

Traps: "coverage depends mainly on number of speakers" (false: commercial, military). "Japanese is high resource" (false: medium). "Arabic is medium" (false: high). "low resource: limited labelled data, some tools" (false: no labelled, no tools).

English-first assumptions

assumption	broken by
whitespace between words	Chinese, Japanese, Thai, Lao, Khmer, Burmese
word order = relations	case marking: Russian, Finnish, Korean
one word = one meaning	agglutinative: Finnish, Turkish, Hungarian; compounding: German, Dutch, Swedish
left to right	Arabic, Hebrew, Urdu, Persian

- **Thai**: spaces end clauses or sentences (no full stop). **Right-to-left** in Unicode: stored left to right, rendered right to left. **Chinese**: verb never marks tense (去 go/went); past from sentence-final particle 了.

Traps: "Thai uses spaces between words" (false: ends clauses). "Turkish is the compounding example" (false: agglutinative). "Unicode stores Arabic right to left" (false: rendered).

Writing systems

type	one symbol =	examples
alphabetic	consonant or vowel; small inventory	Latin, Cyrillic, Hangul
syllabic	whole syllable	Hiragana, Katakana
logographic	word or morpheme; little sound	Hanzi, Kanji
alphasyllabic	consonant + inherent vowel	Devanagari

- **Arabic**: root only, vowels omitted. **Hangul**: letter shapes encode articulatory features. **Script is not language**: Arabic script also Persian, Urdu; Serbian, Kurdish, Uzbek, Punjabi use two scripts. Own script: Armenian, Georgian, Greek.

Traps: "Kanji is syllabic" (false: logographic). "logographs carry much sound information" (false: little).

Code pages and Unicode

- **Code pages** (Latin-1, Shift-JIS, GBK, KOI8-R): ASCII's 128 (or 256) byte codes read per script. **–** wrong decoding: gibberish; one script per document.
- **Unicode**: one universal character set; each character of the set gets one codepoint `U+hex`. Defines meaning, not bytes (the encoding stores it). Superset of ASCII (U+0000 to U+007F), so English-only Latin-1 is compatible. U+0000 to U+10FFFF: 1.1M+ codepoints in 17 planes of 65,536 (slide typo 65,356).
- **Planes**: 0 BMP (most modern languages, most common CJK, Private Use Area); 1 historic and some modern scripts, music, emoji; 2, 3 more CJK; 15, 16 private use.

Traps: "Unicode specifies the bytes" (false: the encoding does). "emoji are in the BMP" (false: plane 1). "all CJK is in the BMP" (false: the most common).

Codepoints and normalisation

- **Codepoint**: abstract number, unit of identity. **Character/grapheme**: what a reader perceives as one unit, one or more codepoints. **Glyph**: stylistic rendering. café: NFC `U+00E9` (4 codepoints) or NFD `e+U+0301` (5); naive comparison fails.
- **Forms**: C composed, D decomposed; NFC, NFD canonical, non-lossy; NFKC, NFKD compatibility, lossy. **NFD**: base + combining marks; keeps semantic and visual equivalence. **NFKD**: most basic parts, splits ligatures; styling not reconstructable (ſ → s, kept by NFC/NFD). **Use** NFD to compare rendered text; NFKD for search indexing, NLP analysis (not necessarily generation).
- **Crawled text** mixes NFC and NFD: dedup, search, string equality, prefix matching go wrong. Fix: normalise consistently (e.g. decompose, then compose).

Traps: "NFD is lossy" (false: NFKD is). "use NFKD to compare rendered text" (false: NFD). "NFC splits into base + marks" (false: NFD). "crawled Unicode is consistently normalised" (false).

UTF-8, UTF-16, UTF-32

- **UTF-8**: 1 to 4 bytes per codepoint, ASCII compatible; web, Linux, NLP tools; most common. **UTF-16**: 2 or 4, not ASCII compatible; Java, JavaScript, Windows. **UTF-32**: exactly 4, not ASCII compatible; rare, easy indexing, memory-hungry.
- **UTF-8 byte count**: up to U+007F 1 byte (7 bits); U+07FF 2 (11); U+FFFF 3 (16); U+10FFFF 4 (21). Beyond the BMP: 4 bytes in UTF-8 and UTF-16. Lead bits: `0` single byte (ASCII); `11` first byte, number of 1s = length; `10` continuation. A 1, é 2, 中 (most CJK) 3, emoji 4.

Traps: "UTF-16 is ASCII compatible" (false). "UTF-16 uses exactly 2 bytes" (false: 2 or 4). "a byte starting 10 begins a sequence" (false: continuation). "most CJK takes 2 bytes in UTF-8" (false: 3).

Bytes, breaking and other pitfalls

- **Byte inflation** (bytes/char): hello 1.0, café 1.25, 中文 3.0. Byte-level tokenisers: more bytes, splitting tends to be more aggressive; can create non-existing characters.
- **Breaking**: bytes can split a codepoint; codepoints can split base from diacritic; graphemes ideal but very language-specific. Mixed strategies misalign spans (SQuAD-style QA) and cross-corpus counts. **Turkish** lowercase(I) = ı. Codepoints not always alphabetical; Swedish å, ä, ö after z; German ä → ae. Punctuation (“” «») hand-mapped.

Traps: "more bytes, less splitting" (false). "codepoint breaking keeps base and diacritic together" (false). "grapheme breaking is language-independent" (false).

L03 Morphology and word formation

Vocabulary, lexicon, short-lists

- **Vocabulary**: word inventory. **Lexicon**: meanings and functions of words. **Dictionary**: written lexicon.
- **WordNet**: hand-crafted (hyponym *car :: vehicle* is-a, meronym *wheel :: car* part-whole); expensive, incomplete, language-specific.
- **Fixed size**: all weight matrices, input embeddings and output layer included; output layer costs significant memory.
- **Short-list**: keep the n **most frequent** words (10K to 200K), map all rare words to `<unk>`. **–** all rare words get **equal probability** in a given context. Words are indices (*move/moved* unrelated); morphology helps unseen words.

Traps: "a lexicon is the word inventory" (false: vocabulary; lexicon = meanings, functions). "a short-list keeps rare words distinct" (false: all become `<unk>`).

Word segmentation: maximum matching

- **Word segmentation / tokenisation**: boundaries where none are written (Chinese, Japanese, Thai, Burmese); **before** POS tagging, parsing, anything else. Ambiguity is **genuine**. **vs subword segmentation:** for rich morphology.
- **FMM** (left to right) / **BMM** (right to left, prepends): precompiled vocabulary V, max word length L; take the **longest** word in V; a single character is **always accepted**. Greedy. **+** simple, strong baseline, fast fallback. **Often disagree** on ambiguous strings: 研究生活, V = {研究, 研究生, 生活, 生, 活}: FMM 研究生 | 活, BMM 研究 | 生活.
- **Bidirectional**: run both; agree = accept with high confidence; disagree = **tie-breaking heuristic**: fewer words or fewer single-char words (word count ties above; single-char picks BMM).
- **Limits** (bidirectional too): labour-intensive vocabulary; new words (names, typos) **cannot be matched**, **break silently** (single characters or mismatched); **no mechanism to exploit data**.

Traps: "FMM and BMM give the same output for the same vocabulary" (false: often disagree). "bidirectional matching uses probabilistic evidence" (false: heuristic; HMM does). "unknown words are flagged" (false: break silently). "maximum matching improves with more data" (false).

BMES tagging with an HMM and Viterbi

- **BMES**: per character. **B** first of multi-char word, **M** interior (words of **3+** chars), **E** last of multi-char word, **S** one-char word. Boundary **after every E and S**. Solvers: HMM, CRF, any neural labeller.
- **HMM**: **hidden** tags generate **observed** characters. Transition $P(t_i \mid t_{i-1})$ (tag bigrams; B never followed by B). Emission $P(c_i \mid t_i)$: **character given tag**. Ideally labelled data (MLE counts); **can** train unsupervised (Baum-Welch, EM), **quality drops**.
- **Viterbi**: dynamic programming (scoring all labellings intractable): $s[i][t] = \max_{t'} s[i-1][t'] \, P(t \mid t') \, P(c_i \mid t)$ = **best** path ending at i with tag t; backpointers. $O(n \times |T|^2)$.
- **HMM vs FMM/BMM**: **+ no fixed dictionary** (characters); ambiguity resolved by **probabilistic evidence**; trainable from **any** segmented corpus, **even another domain**. **–** first-order: tag depends only on **preceding tag and current character** (higher orders possible, more involved); **no longer-range context**: motivates **CRFs**.

Traps: "emission = tag given character" (false: character given tag). "characters are hidden" (false: tags). "Viterbi scores every tag sequence" (false: best path per cell). "$O(n^2 \times |T|)$" (false: $O(n \times |T|^2)$). "HMM needs in-domain data" (false).

Word formation and morphemes

- **Inflection**: **retains** POS, changes number, tense, case, mood, aspect (*see/saw*). **Derivation**: changes POS **and/or** meaning (*happy/happiness*; *perfect/imperfect* keeps POS). **Compounding**: several words into one (*Tischbein* N+N, *sichergehen* adj+verb). **All three very productive** (iterative, new combinations): vocabulary grows a lot. English case **only on pronouns**, Russian on **all** nouns.
- **Morpheme**: smallest meaning-carrying part. **Root**: **free**. **Affix**: **bound**: **prefix** (English: mostly one), **suffix** (can be several), **infix** (*passersby*), **circumfix** (*ge-seh-en*, front and back simultaneously).
- **Concatenative** (all affixes above) **vs non-concatenative**: **apophony** *Buch → Bücher*.

Traps: "derivation always changes the POS" (false: and/or meaning). "compounding only combines nouns" (false: *sichergehen*). "a root is bound" (false: free). "apophony is concatenative" (false).

Morphological typology

type	how	languages
isolating	words tend not to change form; word order, particles	Chinese, Vietnamese
agglutinative	morphemes strung together, one clear meaning each	Turkish, Finnish
fusional	one affix or pattern, several features, often not separable	Russian, Spanish, Arabic
polysynthetic	verb, subject, object, modifiers in one word	Inuktitut, Mohawk

- **Agglutinative vs fusional: separability**. Agglutinative joins may change characters; Russian *-ami* fuses case, number, gender.
- **Word-level NLP breaks**: a lemma can have **thousands** of forms, **massive OOV** with an English-sized vocabulary; analysers **highly language dependent**, so **subword tokenisation**.

Traps: "Arabic is agglutinative" (false: fusional). "Finnish is fusional" (false: agglutinative). "agglutinative morphemes never change form" (false).

L04 Subword segmentation

Why subwords, FSTs and Morfessor

- **Vocabulary**: one of the main bottlenecks of NMT; a word's representation depends on its frequency **and** number of distinct contexts. **Whole words –** huge vocabularies; short-lists map rare words to `<unk>`.
- **Subword**: character n-gram or morphologically meaningful unit (language dependent). Handling rare and OOV words is one main motivation.
- **FST** (historical, hand-built): morpheme in, analysis out. **+** powerful, efficient, cases and exceptions with great accuracy. **–** extremely laborious, totally language dependent, no preference among ambiguous analyses (probabilistic FSTs exist).
- **Morfessor** (Creutz and Lagus 2002): first data-driven; **MDL** = cost of generating the data **plus** cost of the model. **Current methods** (BPE, WordPiece, Unigram LM): frequency-based, no annotation or linguistic knowledge, no labels, strictly concatenative (no character substitutions or deletions). **+** fitted to a desired vocabulary size.

Traps: "FSTs handle exceptions poorly" (false: great accuracy). "MDL minimises model size" (false: model plus data cost). "data-driven segmenters handle character deletions" (false: strictly concatenative).

BPE and WordPiece

- **BPE training**: characters, word boundaries kept (`</w>`); merge the **most frequent** adjacent pair until a **maximum number of merges** (the only hyperparameter; 30K to 128K, high end multilingual). **Greedy** (once merged, stays merged), **deterministic** (one segmentation per word).
- **BPE inference**: characters, then the learned merges **in learned order**. **+** much smaller vocabulary (memory, speed), better translation, far fewer OOVs, for **both** high- and low-resource pairs.
- **Sennrich EN→DE**: best single: char-bigram **ties** joint BPE; best ensemble: char-bigram; per-language BPE single **below** best word-level; syntax-based beats **every** single model, only subword ensembles beat it.
- **WordPiece** (Schuster and Nakajima 2012): **bottom-up** like BPE, and **predates** it. Non-initial characters prefixed `##`; merge the pair with highest $\text{count}(a,b)/(\text{count}(a)\,\text{count}(b))$ until the target vocabulary size. **Inference**: left to right **greedy longest match** (no merge replay); no match at a position: `[UNK]` (L06 table: whole word).

Traps: "at inference BPE merges the input's most frequent pair" (false: replays learned merges). "WordPiece merges the most frequent pair" (false: count ratio).

Unigram LM and DP segmentation

- **Unigram LM** (Kudo 2018, in SentencePiece): considers **all** segmentations, best **global** one, **multiple** (sampling, n-best), **top-down** (large vocabulary shrunk to target).
- **Pruning**: start V = substrings occurring more than once, **not crossing words**. While $|V| > k$: fit θ; $L_t = p_\theta(D) - p_{\theta'}(D)$ (θ' without t); remove $\min(|V| - k, \lfloor \alpha |V| \rfloor)$ tokens with **highest** L_t (exam answer: slide; Kudo's paper: lowest); $\alpha \in [0,1]$ hyperparameter.
- **Unigram inference**: **Viterbi** gives only the best; n-best via **Yen's/Eppstein's** algorithms or disallowing sub-solutions.
- **Brute force**: 2^{n-1} segmentations, each scored. **DP**: solve each sub-problem once, store it: **extra memory to save time**. Top-down memoization and bottom-up: **same asymptotic** run time. $m[i,j] = p$ (whole span), replaced by $m[i,k] \cdot m[k+1,j]$ (**product**; segments independent) if larger.

Traps: "starts from single characters" (false: substrings occurring more than once). "all surplus tokens removed at once" (false: at most $\lfloor \alpha |V| \rfloor$ per round, refit between). "Viterbi returns n-best" (false). "bottom-up DP is asymptotically faster" (false: same). "DP saves time and memory" (false: spends memory).

SentencePiece, OOV and byte-level BPE

- **SentencePiece**: a **toolkit** (BPE **and** Unigram LM). Standard BPE assumes word boundary detection; SentencePiece reads a **raw Unicode stream including spaces**, space escaped as _. **+ lossless** desegmentation, languages without word boundaries.
- **OOV**: BPE and Unigram LM **very low**, not zero: typos **sometimes**, unknown characters **will**.
- **Byte-level BPE**: BPE on **UTF-8 bytes**, base alphabet fixed at **256**: **zero OOV guaranteed**, even unseen scripts. **–** non-Latin, non-ASCII scripts: more bytes per character, extra merges, **shorter** subsegments.

Traps: "BPE and Unigram LM guarantee zero OOV" (false: byte-level only). "byte-level BPE works on Unicode characters" (false: bytes). "SentencePiece is only Unigram LM" (false: BPE too). "non-ASCII scripts get longer subsegments" (false: shorter).

BPE, WordPiece, Unigram LM compared

	BPE	WordPiece	Unigram LM
builds	merges up	merges up	prunes
criterion	pair count	count ratio	likelihood
new word	replay	longest match	Viterbi/sample
fallback (L06)	char/byte	word → `[UNK]`	char
marker	`@@`, `</w>`	`##`	_ = space
output	one	one	one or sampled
used in	GPT, RoBERTa, XLM	BERT, mBERT	T5, XLM-R, ALBERT

BPE vs Unigram LM (Bostrom and Durrett 2020)

- **Segments**: Unigram LM closer to morphemes, **longer** (English **and** Japanese), more moderate-frequency tokens (English); BPE more word-initial capitals (_H), Unigram LM more suffixes (*s, ing, ed*); in Japanese BPE attaches particles to words.
- **Results**: vs gold segmentations Unigram LM wins precision, recall **and** F1 in **both**, gap **much larger in English**. Downstream ahead on **every** task: **slightly** in English, **a lot** in Japanese.

Fertility

- **Fertility**: average segments per word; depends on merges (BPE) or vocabulary size (Unigram LM), word length, morphological richness. Isolating, Latin-script: **typically close to 1** for common words. No scheme dominates across all languages and scripts. **High fertility –** harder to reassemble meaning, **pays more** under per-token pricing, context fills **faster**.

Traps: "isolating languages have high fertility" (false: near 1). "high-fertility languages fill context more slowly" (false: faster).

L05 Static embeddings

Count vectors and embeddings

- **Count vectors** (distributional semantics): context-word counts in a window (or sentence, document); context and target vocabularies may be the same; cosine. High-dimensional, discrete, sparse. **+** unsupervised; interpretable dimensions. **–** sparse despite stemming; no signal vs noise criterion.
- **Embeddings**: low-dimensional, dense, continuous, learned by a prediction task, usually as a by-product. **Representation learning**: embeddings are the main objective.

Traps: "embeddings are high-dimensional and sparse" (false: count vectors).

Word2Vec: CBOW and skip-gram

- **CBOW**: centre word from context; $h = \frac{1}{2n} W w_C$ (average; figure says SUM); one hidden layer, **no non-linearities**, simpler than PNLM; $n \in \{2, 5, 10\}$.
- **Both**: $\text{softmax}(W'h)$. W' **not necessarily** $W^\top$. Embedding: typically W, or $W^\top + W'$.

	CBOW	skip-gram
predicts	centre from context	each context word from centre
hidden	average of context	centre embedding Ww_I
outputs	one	$2n$, shared W'
assumes	order ignored	context words independent

Traps: "skip-gram predicts the centre word" (false: CBOW). "CBOW has a non-linearity" (false). "Word2Vec requires $W' = W^\top$" (false). "embeddings only from W'" (false: typically W).

Negative sampling

- **Negative sampling**: full output layer too costly, so classify whether a word occurs in the input word's context: **1 positive** + k **negatives** (same w, noise word as c); $p(D{=}1) = \sigma(v_c \cdot v_w)$.
- **Parameters**: k = **5 to 20**; unigram$^{0.75}$ (biases towards **rarer** words); context **typically 2 to 5**; **more data → smaller** context, fewer negatives.

Traps: "negatives drawn uniformly" (false). "0.75 shifts mass to frequent words" (false: to rarer). "more data needs more negatives" (false). "negative sampling uses a softmax over the vocabulary" (false: sigmoid of dot product).

GloVe

- **GloVe**: combines global co-occurrence statistics (count-based, tends to do worse) with local prediction-based updating (each window an independent event, still tends to win).
- **Objective**: $\sum f(X_{ij})(w_i \cdot \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij})^2$, weighted least squares on log counts. X built in **one pass**; training sees only X. **Final vector** $w_i + \tilde{w}_i$.
- f: $(x/x_{max})^\alpha$ below x_{max}, else 1; $x_{max} = 100$, $\alpha = 0.75$. De-emphasises very rare (noisy) **and** very frequent pairs: rises from zero, **caps at 1**.

Traps: "GloVe treats windows as independent events" (false: Word2Vec). "count-based outperforms prediction-based" (false). "f gives frequent pairs the lowest weight" (false: capped at 1). "GloVe subsamples frequent words" (false: Word2Vec; GloVe uses f).

Evaluating embeddings

- **Intrinsic**: embeddings alone; **extrinsic**: downstream (MT, IR). **Qualitative**: examples; **quantitative**: overall score. **Word similarity**: **Spearman** with human rankings. **WS-353 = relatedness**, **SimLex-999 = similarity**. **Analogies**: **accuracy**, incl. **syntactic**. **Visualisation** (PCA; t-SNE hyperparameters matter a lot): **complements, does not substitute** quantitative or extrinsic evaluation.

Traps: "SimLex-999 annotates relatedness" (false: WS-353). "similarity scored by accuracy" (false: Spearman).

Side 2 · L05 fastText **1** · Crosslingual embeddings **1** · Supervised mapping: Procrustes, orthogonal **1** · Hubs and CSLS **1** · Unsupervised alignment: MUSE **1** · Where unsupervised fails (Søgaard et al. 2018) **1** L06 Static and contextual representations **1** · Pre-training and fine-tuning **2** · BERT pre-training **2** · BERT fine-tuning and whole-word masking **2** · mBERT and language sampling **2** · Zero-shot crosslingual transfer findings **2** · XLM-R **3** L07 Multilingual vs crosslingual, parallel data **3** · Pretrained models compared **3** · LASER **3** · XLM: CLM, MLM, TLM **3** · InfoXLM **3** · Within vs across languages **3** · Seq2seq and NMT **4** · Parent-child transfer **4** · Multilingual NMT **4** · BART **4** · mBART **4**

L05 Static embeddings (cont.)

fastText

- **fastText**: remedies one opaque vector per word type (morphologically rich languages suffer most, no OOV fallback): skip-gram with negative sampling; **centre word = sum of character n-gram vectors** (n default **3 to 6**, < > boundaries) **plus whole word** <w>. **Context vectors stay per word.** Same gradient to every n-gram. **Returns the n-gram vectors**; **OOV words get vectors.**
- **Results**: similarity best or tied except **English WS-353** (CBOW wins); gains in morphologically rich languages and English rare words. Analogies: gains on **syntactic**; **semantic** barely change, drop for German.

Traps: "context words also built from n-grams" (false). "cannot represent unseen words" (false). "largest gains on semantic analogies" (false: syntactic).

Crosslingual embeddings

- **Shared space hypothesis**: train separately, jointly or on aligned data, alignment needed either way. Similar relational structure, **approximate isomorphism** (same cloud shape; orientation, scale may differ). Only a hypothesis. If true, **one general linear map** aligns them (linear because analogies are already linear in one space).
- **Supervision**: **supervised** thousands of seed pairs (top frequent words; words only, no phrases); **semi-supervised** hundreds, bootstraps by confidence; **unsupervised** none, adversarial.

Traps: "spaces differ only by a rotation" (false: general linear map). "semi-supervised uses thousands" (false: hundreds).

Supervised mapping: Procrustes, orthogonal

- **Mikolov et al. (2013)**: $\min_W \sum_i \|Wx_i - y_i\|^2$, **Procrustes** (exam answer: slide; Conneau: only the orthogonal SVD one); exact closed form exists, used **SGD** (scales better, memory-friendlier). Worse for rarer words. **–** stretches, shears; overfits the seed dictionary; Euclidean training vs cosine retrieval (equal only for unit vectors).
- **Xing et al. (2015)**: **orthogonal** $W^\top W = I$, unit vectors; rotations and reflections only, preserves distances and angles; exact by **SVD**. Better at every dimension; unconstrained **worsens 300 to 700 dims** (equal dims), orthogonal slightly improves.

Traps: "orthogonal maps can rescale" (false). "unconstrained improves with dimension" (false, equal dims). "orthogonal needs gradient descent" (false: SVD).

Hubs and CSLS

- **Hubness**: few vectors are NN of many points regardless of similarity; high-dimensional geometry, **not specific to embeddings**, **worse with more dimensions**.
- **CSLS**: $2\cos(x,y) - r_T(x) - r_S(y)$; $r_T(x)$ = mean cosine of x to its K NN in the **target** space, $r_S(y)$ in the **source** space. **Penalises** hubs; **no parameter tuning**; beats plain NN on every pair.

Traps: "CSLS boosts candidates near many points" (false). "r_T uses the source space" (false).

Unsupervised alignment: MUSE

- **Generator**: W, one $d \times d$ **linear** matrix, deliberately (a deep one could scramble which word maps where); starts as identity (or random orthogonal); **re-orthogonalised** after each update.
- **Discriminator**: MLP, label smoothing; **50k most frequent** words; alternating steps, other side frozen; generator uses **flipped labels**.
- **Model selection**: not adversarial loss; mean cosine of CSLS NN of **10k most frequent** source words. **Refinement** ("semi-supervised" on slide, no human dictionary): **mutual** CSLS NN among frequent words, SVD, repeat.
- **Results**: on par or better than supervised for English-Spanish, -French, -German; well short for -Russian, -Chinese.

Traps: "generator is a deep network" (false). "checkpoint by discriminator loss" (false). "refinement needs a seed dictionary" (false). "always worse than supervised" (false).

Where unsupervised fails (Søgaard et al. 2018)

- **Marking** of grammar: on dependent (possessor), on head (verb), both (double), neither (zero).
- **Fails**: English-**Estonian, Finnish, Greek** (mixed/double marking): grammar spread over many rarer forms, **non-isomorphic**. **Identity baseline** (identically spelled words): beats adversarial on **every English pair**; **Estonian-Finnish** is the exception. **Not data size**: far more Finnish data, still zero.
- **Domain mismatch**: adversarial fails completely, even English-Spanish; identity degrades but survives. **Model**: window, n-gram changes barely matter; **skip-gram on one side, CBOW on the other fails**.
- **Eigenvector similarity**: NN graph; Laplacian $A - D$ (exam answer: slide; standard $D - A$); largest $n\%$ eigenvalues (bigger = denser connectivity); Δ = sum of squared differences. Bigger Δ = less isomorphic, no rotation fixes it; no alignment needed. **One direction only**: low Δ does not prove isomorphism. Highest for the failing pairs.

Traps: "a bigger Finnish corpus fixes it" (false). "fails for every Finnish pair" (false). "identity beats adversarial on every pair" (false: every English pair). "low Δ shows isomorphism" (false). "window size breaks alignment" (false: skip-gram/CBOW mismatch does).

L06 Contextual embeddings

Static and contextual representations

- **Static embedding**: one vector per word type. Static from Word2Vec (limited window) and also as a **by-product** of a deep task model (classification, MT) reading the whole sentence.
- **Contextual representation**: the hidden states above a word, different per occurrence. **Balance**: enough context to tell occurrences apart, not so much that the word's own contribution gets unclear. **Strategies**: right layer(s), right task(s), residual layers.

Traps: "embeddings learned inside an MT model are contextual" (false: the embedding layer is static). "more context always gives a better word representation" (false: the word's contribution gets unclear).

Pre-training and fine-tuning

- **Pre-training**: model of wider context (LSTM, Transformer **or CNN**) on a general task with very large data (word prediction); updates θ. **Fine-tuning**: adds task model θ' for the real task; updates θ' **and maybe** θ. **Feat** (feature extraction): encoder frozen, only the head trained.
- **Encoder-decoder**: original Transformer (MT, summarisation, QA). **Encoder-only**: predicts its own noised input (purest form: auto-encoder), **BERT**. **Decoder-only**: next-word prediction, **all** current LLMs; can frame encoder-decoder tasks.

Traps: "fine-tuning always updates θ" (false: θ' always, θ maybe). "MT requires an encoder-decoder" (false: decoder-only can frame it). "contextual pre-training requires a Transformer" (false: LSTM, CNN too).

BERT pre-training

- **BERT**: Transformer **encoder**. Pre-training **self-supervised** (missing words), task relevance **not required**; fine-tuning **supervised**, works with small data. Input = token + segment (E_A/E_B) + position embeddings.
- **MLM**: tokens **selected** with $p_{mask} = 0.15$; of those **80%** `[MASK]`, **10%** random, **10%** unchanged. **Loss at all selected positions**, only there. Masks drawn once in pre-processing (**static**).
- `[CLS]`: not tied to a word, **never masked**, represents the **entire** input; loss from a classification task (NSP). **NSP**: **50%** B follows A, **50%** B random **from the corpus**.
- **Sizes**: base 110M, 12 layers, 768; large 340M, 24 layers, 1024; **16 heads both** (exam answer: slide; paper: 12 for base). BookCorpus + English Wikipedia, WordPiece.

Traps: "15% of tokens are replaced by `[MASK]`" (false: 80% of the 15%). "loss only at `[MASK]` positions" (false: all selected). "`[CLS]` is masked like other tokens" (false). "NSP negatives use the preceding sentence" (false: random).

BERT fine-tuning and whole-word masking

- **Fine-tuning head**: small-ish, **randomly initialised**; word tasks on **top encoder layers**, classification on the `[CLS]` **top layer**; BERT updated or not. **2 to 4 epochs**.
- **NLI**: entails / contradicts / no relationship, 3-way. **XNLI**: crowd-sourced test/dev pairs **translated** into 14 languages. **SQuAD**: **start/end span**; **one** start vector S, **one** end E, span maximises $T_i \cdot S + T_j \cdot E$, $i \le j$.
- **Whole-word masking**: mask all pieces of a word; **rate unchanged**. Pieces carrying **morphology** (tense, number, case) are **easier** to predict than stems. Helps BERT-Large, cased and uncased.

Traps: "head initialised from the pre-trained model" (false: random). "whole-word masking raises the masking rate" (false). "a start/end vector per position" (false: one shared). "suffix pieces are harder than stems" (false: easier).

mBERT and language sampling

- **mBERT**: **Same architecture and loss** as BERT (MLM + NSP), 104 concatenated Wikipedias. NSP pairs **within one language**, no mixing in a sequence; a **batch can** mix languages. **Lacks**: crosslingual loss, parallel data (none intentionally), language-ID embedding, language-specific parameters (adapters), alignment loss.
- **Shared vocabulary**: WordPiece (L04) **~120k** (English BERT 30k); BPE or Unigram LM work **similarly well**. ~178M vs 110M parameters: **only the vocabulary**. Chinese, kanji, **hanja** split into characters first. The α smoothing (below) also applies to subword frequencies; still favours **high-resource and Latin script**; low-resource, morphologically rich: **higher fertility**. **Uncased** variant lowercases and **strips accents** (hurts diacritic languages).
- **Sampling** $p_i = n_i^\alpha / \sum_{i'} n_{i'}^\alpha$, n_i = resource size, α = importance (smoothing; $1/T$ in L07).

α	effect
1	proportional: English, large Wikipedias dominate
0.7	up-weights low-resource, down-weights high
0	uniform: tiny Wikipedias oversampled, memorised

Traps: "mBERT has a language-ID embedding" (false). "NSP pairs mix languages" (false). "more parameters because more layers" (false: vocabulary). "$\alpha = 0$ is proportional" (false: uniform). "$\alpha = 0.7$ down-weights low-resource" (false).

Zero-shot crosslingual transfer findings

- **Zero-shot**: task data in A improves the task in B with **no** task data in B.
- **Pires 2019** (NER, POS, 16 languages): zero-shot decent, below in-language, **asymmetric** (en→nl beats nl→en). **Cross-script works**; pairs with Japanese drop (maybe typology). Jaccard overlap of **named-entity** wordpieces: English BERT rises with it, **mBERT high even at low overlap**. From SVO, SVO targets beat SOV; more shared WALS features, better POS. **Translation retrieval**: **peaks in middle layers**.
- **Wu and Dredze 2019** (from English): XNLI translate-train **beats** zero-shot mBERT, which loses even to XLM MLM-only; NER **beats** Xie et al.; POS **below** Kim et al. **Freezing** lowest k layers (k = 0 is embeddings): $k \le 6$ mostly neutral or **helps**; $k = 9$ hurts NLI, NER, POS (MLDoc gains); no fine-tuning (Feat) below best fine-tuned. **Overlap vs zero-shot** per task: NER near-perfect correlation, XNLI **none**, others positive.
- **K et al. 2020** (bilingual BERT, XNLI): **fake English** (zero overlap) costs **little**: overlap not the main cause. **Permuting words** in pre-training only: significant drop, transfer **not eliminated**. **Depth** has the largest architecture impact (parameters ~constant): in-language saturates, zero-shot keeps rising. **Mixed-language** premise/hypothesis: below **both** monolingual pairs.
- **Rajaee and Monz 2024** (mBERT, XNLI 15×15): **across < within on average for every language** (single English-hypothesis cells can beat within); same-language premise best per hypothesis language; foreign premise + **English hypothesis** beats the reverse. **Hypothesis**: transfer of **heuristics can contribute**.
- **What drives transfer**: shared vocabulary **plays a role, not the main cause**; structure matters (permutation hurts, does not eliminate); deeper = **more language-independent**; **no single factor**.

Traps: "mBERT transfers poorly across scripts" (false). "retrieval best at the top layer" (false: middle). "removing subword overlap eliminates transfer" (false: small cost). "heuristics are the main cause" (false: hypothesis that they can contribute). "shared vocabulary plays no role" (false).

XLM-R

- **XLM-R** (Conneau 2020): RoBERTa-style mBERT. **Drops NSP** (little use, sometimes hurts): MLM only. **Dynamic masking** (new mask each pass) **vs static** (once, in pre-processing). **CC-100** (filtered CommonCrawl, **100** languages, ~2 orders of magnitude more data). **250k** Unigram LM (SentencePiece), double mBERT's, better fertility. Still **no crosslingual signal**.
- **XLM-R vs mBERT** within and across: see L07.

Traps: "XLM-R covers more languages" (false: 100 vs 104). "XLM-R keeps NSP" (false: MLM only).

L07 Crosslingual NLP

Multilingual vs crosslingual, parallel data

- **Multilingual model** (mBERT, XLM-R): one model and subword vocabulary, MLM context **within the same language**. **Nothing explicitly links languages**: predicting W in L gains nothing from unrelated context in L'. mBERT sees no translation pair; alignment is an **emergent side effect**.
- **Crosslingual knowledge transfer**: fine-tune on a task in A, test in B. Zero-shot results **seem to support** it emerging without crosslingual signal; crosslingual tasks **cast doubt**.
- **Parallel data**: **meaning-equivalent** sentence pairs. **vs comparable data:** same **topic**, not fully parallel, a **sliding scale** (news, Wikipedia).
- **Sources**: UN, EU, news, **OPUS**; crawl (**document, then sentence alignment**, or straight from Common Crawl). Equivalence: dictionary overlap, relative length (old); **sentence embeddings** (recent). **vecalign**: LASER matching at massive scale.

Traps: "joint training on many languages gives an explicit crosslingual signal" (false: MLM stays in one language). "comparable data is translation pairs" (false: parallel).

Pretrained models compared

model	data	objective	vocab; language ID
mBERT	Wikipedia	MLM + NSP	WordPiece ~120k
XLM-R	CC-100	MLM	Unigram 250k
LASER	parallel	**translate**	BPE; L_{id} to decoder
XLM	Wiki + parallel	MLM or CLM + **TLM**	BPE; lang. embedding
InfoXLM	CC-100 + parallel	MMLM + **TLM** + **XLCo**	XLM-R's
BART	English	denoising	
mBART	monolingual, many	denoising	ID token <En>

- **Bold** = crosslingual signal; no language ID unless listed. XLM from scratch; InfoXLM from XLM-R.

LASER

- **LASER**: encoder-decoder trained to translate on parallel data. **Encoder**: shared BPE embeddings, **BiLSTM** stack, **max pooling** into one sentence embedding; never told the input language. **Decoder** (LSTM): sentence embedding + previous token + **language ID** L_{id} of the output; later **discarded**.

Traps: "LASER's encoder gets the input language ID" (false: decoder, output language). "LASER uses `[CLS]`" (false: max pooling).

XLM: CLM, MLM, TLM

- **CLM**: next word from prefix. **MLM**: masked word from context (BERT). **TLM**: MLM's loss on sentence + translation concatenated, masks in **both** halves, so a masked word can be read off its translation. **Positions restart at 0** in the second sentence; **language embeddings** mark each half.
- **TLM is the crosslingual objective**, always paired: **MLM + TLM** or **CLM + TLM**. **NSP dropped**.
- **XNLI**: **translate-train** (MT training set), **translate-test** (MT test into English), **zero-shot** (English only). MLM + TLM beats MLM in **every** language (least for English). Zero-shot beats translate-test **on average only** (translate-test wins every European language, tr, ar). **Translate-train best on average**.
- **Nepali LM** gains more from **Hindi** (related) than English; Hindi + English together best.

Traps: "XLM keeps NSP" (false). "TLM is used alone" (false: with MLM or CLM). "TLM positions continue from the first sentence" (false: restart at 0). "zero-shot beats translate-test in every language" (false: on average).

InfoXLM

- **InfoXLM**: **initialised from XLM-R** plus XLM's parallel-data ideas; three losses, **equally weighted**. **MMLM**: like MLM, but instead of directly optimising the ground-truth loss uses **negative sampling** (exam answer: slide; paper: plain MLM). **TLM** as in XLM, but **no language embeddings**. **XLCo**: `[CLS]` vectors of a parallel pair, **classify whether parallel**, negatives = **random sentences**.
- **TLM vs XLCo:** TLM aligns **tokens**, XLCo **sentences**; XLM is token level only.

Traps: "InfoXLM is trained from scratch" (false). "InfoXLM uses language embeddings" (false). "XLCo predicts masked words from the translation" (false: TLM).

Within vs across languages

- **Within vs across**: fine-tuned on English; input parts in one language or two. **Every model lower across**. XLM-R beats mBERT within on both tasks (on average); across, better on XNLI but **worse on QA** (works mainly with an **English question**). **InfoXLM**: slightly better within, much better across, **smallest gap**, biggest gain on QA. Grids **not symmetric**.

Traps: "XLM-R beats mBERT across languages on both tasks" (false: worse on QA). "InfoXLM gains mainly within a language" (false: across).

Seq2seq and NMT

- **Seq2seq vs sequence labelling**: **no isomorphic** x_t, y_t relation; lengths **may differ**. $p(y_t \mid Y_{<t}, X)$: X = encoder output, $Y_{<t}$ = decoder prefix.
- **Sutskever 2014**: LSTM encoder and decoder, decoder initialised with the **last** encoder state: **information bottleneck**; Transformer's **source-target context** layer (cross-attention) removes it.
- **NMT quality**: data **amount, variation, relevance**. **86%** of directions poor (limited parallel data); generalises not very well beyond training data, **not at all** beyond specific directions. Small data: NMT **far below** phrase-based, ahead only when large.

Traps: "seq2seq assumes equal lengths" (false). "Sutskever's decoder attends to every encoder state" (false: last state).

Parent-child transfer

- **Transfer learning** = basically **fine-tuning**: big pair (**parent**), then continue on low-resource pair (**child**). **Works best if**: **identical target**, **related source**, **decoder frozen**, shared vocabulary.
- **Zoph**: **not shared**: source vocabulary, source embeddings; **shared**: target language, all else; freeze target-specific.
- **Freezing**: all frozen = zero; **source embeddings** biggest jump; source RNN, target RNN, attention keep helping; unfreezing **target embeddings hurts** (slide's "decoder frozen").
- **Relatedness**: French parent beats German for a Spanish child; reshuffled-vocabulary French still gains: **structure transfers**. **Kocmi and Bojar**: shared vocabulary, **unrelated** parent helps as much, even with a different target. Reversed (small parent): little or **hurts**. **Aji**: **inner layers** carry most; **embeddings alone worse than random init** on average.
- **Fine-tuning** (sample exam, no official key): starts from general-domain parameters; more in-domain updates help, then hurt; suffers catastrophic forgetting.

Traps: "Zoph's parent and child share source embeddings" (false). "unfreezing target embeddings helps" (false). "with a shared vocabulary only a related parent helps" (false).

Multilingual NMT

- **Johnson 2017**: **one system** for all directions; transfer **in parallel** (parent-child: in stages). **English-centric** (xx→en, en→xx). **Target language tag** (<2es>) prepended to the source. **All parameters shared**; one WordPiece vocabulary; source, target, output embeddings shared; directions mixed within a batch.
- **Many-to-one**, **one-to-many**: training; **many-to-many: testing only** (zero-shot). Vs bilingual: many-to-one better everywhere (small); one-to-many **mixed**. **Signs of transfer**: zero-shot quality; beating the single-direction baseline.
- **Transfer** (better from sharing) **vs interference** (worse from competing for capacity). **Arivazhagan**: low-resource gain **far more into English** than out; high-resource **lose** (no multilingual model beats bilingual); **All→All** weakest multilingual everywhere.
- **Sampling**: proportional (T=1): high-resource dominate; uniform (T=100): they suffer; best: over-sample low, under-sample high. $p_i \propto n_i^{1/T}$ (L06's $\alpha = 1/T$):

T	high	medium	low
1	best	worst	worst, below bilingual
5	middle	best	near T=100
100	worst	middle	best

- **Zero-shot**: entirely transfer; more training languages help **in most cases**; **still lagging**.

Traps: "the tag names the source" (false: target). "uniform sampling hurts low-resource" (false: high). "higher T moves towards proportional" (false: uniform).

BART

- **BART**: **bidirectional encoder** reads corrupted text, **autoregressive decoder** rebuilds the original; monolingual data. **vs BERT** (encoder only), **GPT** (decoder only).

noise	source	learn
token masking	→ `[MASK]`	tokens
token deletion	removed, unmarked	tokens + **where**
text infilling	span → **one** `[MASK]`	span + **how many**
sentence permutation	shuffled	order
document rotation (slide: permutation)	random start	real start

- **Infilling best** single noise on almost every task; deletion beats masking on generation; rotation, permutation alone poor.
- **Fine-tuning**: classification from the **last hidden time step**; span: **begin and end** labels; MT: random-init **source encoder replaces the embedding layer** of BART's encoder (**frozen or updated**; source vocabulary may differ). **Frozen slightly below** baseline, **tuned above**.

Traps: "infilling uses one `[MASK]` per token" (false: per span). "BART's decoder is bidirectional" (false). "the new encoder replaces BART's whole encoder" (false: embedding layer).

mBART

- **mBART**: **concatenated monolingual** data; BART objective **within the same language**, **no crosslingual signal**. **Language ID token ends the encoder input, starts the decoder input**. **Fine-tuned directly** on parallel data.
- **vs random init**: better on **every** pair, both directions; gains **largest** low-to-medium; **near zero** with almost no data (10K pairs).

Traps: "mBART is pretrained on parallel data" (false). "mBART needs a new encoder for MT" (false: BART does). "mBART rescues pairs with almost no data" (false).